

1. Identify whether the given expressions are balanced or not

(i) $[9 * (K + 8 - \{6 + e\} + 77) / 100$

* Opening brackets: $[, (, \{$

* Closing brackets: $\},),]$

* The closing square bracket $]$ is missing

Answer: Not Balanced.

(ii) $(((a * b) + [4/2] - a)$

* Opening brackets: $(, (, (, [$

* Closing brackets: $),],]$

* One opening parenthesis $($ is not closed.

Answer: Not Balanced.

2. Compressing a stack with repeated elements

Approach:- Instead of storing duplicate elements separately, store each element as a pair. (value, frequency)

Example:

original stack:

TOP
5
5
5
3
3
2
Bottom

compressed stack:

TOP
(5, 3)
(3, 2)
(2, 1)
Bottom

modified stack ADT operations

push(x):

* If top element = x, increases its frequency.

* Otherwise, push(x, 1).

pop():

* If frequency > 1, decrease frequency

* Otherwise, remove the top pair

peek():

* Return the value of the top pair.

Time complexity:

* push : $O(1)$

* pop : $O(1)$

* peek : $O(1)$

Space complexity

* Best case (many repeated elements): $O(k)$, where

k = no. of distinct consecutive values

worst case (all elements different) : $O(n)$

Trade-off

Advantages:

* saves memory when many consecutive duplicate elements exist

* Stack operations remain $O(1)$.

Disadvantages

* Each stack node stores an extra frequency field.

* No space savings if all elements are unique.